

In-gel protein staining with silver nitrate

Risk assessment

- Formaldehyde is a **KNOWN CARCINOGEN!**
 - Silver nitrate is a **STRONG OXIDANT** that reacts violently with combustible and reducing materials such as alkynes, alkalis, or halides
- ▷ Wear gloves, safety glasses, lab coat
- ☐ Collect formaldehyde solutions as HAZARDOUS WASTE
 - ☐ Collect silver containing solutions as SILVER WASTE
 - ☐ Precipitate silver nitrate solutions immediately with sodium chloride or ascorbic acid to prevent formation of explosive silver azide
 - ☐ DO NOT wash into sewer



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This is a bench card. Full protocol available online.

>> Protein fixation

- ☐ R0170 Fixation solution, 200 mL
- ☐ 30% Ethanol, 400 mL

- (1.) Fix the gel in 100–200 mL fixation solution for 30 min on a rocking table.

⌚ 30 min

Critical: Do not use aldehydes if mass spectrometry is planned. Fixation with methanol and acetic acid is sufficient but slightly less sensitive.



- (2.) Wash the gel twice in 30% ethanol for 10 min each.

⌚ 40 min

- (3.) Wash the gel twice in water for 10 min each.

⌚ 20 min

>> Sensitization and impregnation with silver nitrate

- ☐ R0126 Sodium thiosulfate sensitizing solution, 200 mL
- ☐ R0128 0.2% Silver nitrate, 250 mL
- ☐ R0127 Tetrathionate sensitizing solution, 200 mL

- (1.) Prepare four containers: one for the sensitization solution, two for rinsing with water, and one for the silver nitrate impregnation. Gels should move sequentially through each solution without delay.

- (2.) Sensitize by soaking one gel at a time:

- For a fast, sensitive staining, incubate for 1 min in *sodium thiosulfate* sensitizing solution.
- For consistent, linear staining, incubate for 45 min in *tetrathionate* sensitizing solution.

- (3.) Rinse the gel twice in water for 1 min each. Proceed immediately to silver impregnation.

- (4.) Impregnate with silver nitrate for 20–120 min.



>> Image development

- ☐ R0129 Silver nitrate developer, 200 mL
- ☐ 2% Acetic acid, 200 mL

- (1.) Wash one gel at a time by briefly dipping in water.



- (2.) Transfer gel into developer. Shake until precipitate dissolves.



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Critical: A brown or gray precipitate typically appears within seconds. Shake immediately to redissolve it or risk particulate background.

- (3.) When desired degree of staining is achieved, stop the reaction by transferring gel into 2% acetic acid. Incubate for at least 30 min.

Quality assurance: Intense bands or spots may appear a few minutes into development. Allow development to proceed until background is acceptable. No further changes will occur beyond 45 min. Aim to stop development when protein bands are clearly visible against a faint, uniform yellow-to-brown background. If the background darkens to a deep brown before bands are clearly resolved, the sensitization or impregnation step may need adjustment.

- (4.) Wash gels at least twice for 30 min in water.

Storage of stained gels

- (1.) Leave stained gels in water for up to several days.

Critical: For mass spectrometry, staining, washing, spot excision, and destaining should be performed on the same day.

- (2.) For permanent storage, soak the gel in 10% glycerol for at least 20 min.

Destaining for mass spectrometry

☐  R0130 5% Potassium ferricyanide, 50 mL ☐  R0171 Silver nitrate destaining solution, 10 mL

- (1.) Excise the band or spot of interest from the gel.
- (2.) Cover the gel slice with 150 μ L destaining solution and incubate for 5–10 min. Remove the solution.
- (3.) Rinse the gel slice five times with 150 μ L water for 5 min each. Remove the water between rinses.
- (4.) Soak the gel slice in 200 mM ammonium hydrogencarbonate solution for 20 min.
- (5.) Rinse again with water as before.
- (6.) Proceed to in-gel digestion, or dry and store gel pieces at -20°C until use.

 Recipe (available online)  Troubleshooting (available online)  Notes (available online)

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